Rahul Vigneswaran

Research Fellow
Microsoft Research
Silver Medalist

Last Updated: November 1, 2026
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Education

Indian Institute of Technology, Hyderabad — India
Master of Technology in Computer Science & Engineering (By Research) - CGPA: 9.76/10 — *Rank 1*

Publications

- **IJCAI (W) 2026**
  Rahul Vigneswaran, Divya, Digvijay, Chandana, Vineeth N Balasubramanian.
  Class Incremental Learning Free of Unrealistic Assumptions.

- **TMLR 2026 | AAAI (W) 2026** | Paper |
  Rahul Vigneswaran, Chandana, Vineeth N Balasubramanian.
  Transitioning Heads Conundrum: The Hidden Bottleneck in Long-Tailed Class-Incremental Learning.

- **TMLR 2025** | Paper | Code |
  Rahul Vigneswaran*, Kancheti Sai Srinivas*, Bamdev Mishra, Vineeth N Balasubramanian.
  HARE: Human-in-the-Loop Algorithmic Recourse.

- **ICVGIP 2021** | Paper | Code | Video |
  Rahul Vigneswaran, Marc T Law, Vineeth N Balasubramanian, Makarand Tapaswi.
  Feature Generation for Long-tailed Classification.

- **AAAI 2021** | Paper |
  Adepu Ravi Shankar*, Yash Khasbage*, **Rahul Vigneswaran**, Vineeth N Balasubramanian.
  A Deeper Look at the Hessian Eigen Spectrum of Deep Neural Networks and its Applications to Regularization.

Research Experience

Research Fellow — Microsoft Research, India
Manager: Akshay Nambi — *Present*

- Developing a self-improving LLM pipeline that generates and filters high-quality synthetic trajectories to advance long-horizon reasoning and tool use.

Applied Scientist Intern — Amazon, India
Managers: Promod Yenigalla, Anish Nediyanchath, Anubhav Shrimal — *Mar'25 - Aug'25*

- Developed a new capability for Amazon's internal Agentic assistant (SAPIEN), enabling insight exploration across databases and user files through automatic table identification (**89% accuracy**), context-aware clarification handling, and workflow initiation; demoed to leadership.
- Reduced manual testing effort of SAPIEN (**7**×) by building tools like *Golden Dataset Generator* and *Gamma Testing Framework*, ensuring scalable reliability across use cases.
- Secured a **Top-3 finish** (only intern to do so) at Amazon's internal Hackathon with Promptinator-3000, an automated prompt and dataset generation framework, which achieved **21**× **efficiency gain**, earning recognition from leadership.

Research Assistant — IIT Hyderabad, India
Advisor: Dr Vineeth NB — *Jan'23 - Dec'25*

- Proposed a new realistic setup in Continual Learning that is free of assumption and developed a novel method using adapters to tackle it, which yields **+8% performance boost**.
- Developed a novel method to tackle Transitioning Head problem in Long-Tailed Class Incremental Learning via Early Knowledge Transfer, achieving state-of-the-art results (**+5% performance**).
- Created a human-in-the-loop recourse framework that integrates user feedback, generating personalized counterfactuals and enhancing user satisfaction and transparency. Achieved a **13× improvement** in metrics compared to baselines.

★ Published: **AAAI'26 (W)**, **TMLR'26**, **TMLR'25**. Under Review: **ECCV'26**

Research Intern IIT Hyderabad, India

Advisors: Dr Vineeth NB (IIT-H) & Dr Makarand Tapaswi (IIIT-H) *July'19 - Jan'23*

- Developed TailCalibX, a feature generation technique for Long-Tailed classification that uses calibrated distributions to boost performance on imbalanced datasets, setting a new state-of-the-art.
- Created a Hessian-based regularization method that improves generalization by leveraging the similarity between layerwise and overall Hessians, enhancing regularizer efficiency.

★ Published: **AAAI'21**, **ICVGIP'21**.

Research Projects

Structure Thinking for Enhanced Reasoning in LLMs *March'25 - Aug'25*

Advisor : Anubhav Shrimal (Amazon)

- Introduced a structured thinking framework for reasoning models, achieving **13.9% improvement in pass@k accuracy over chain-of-thought** on complex tool-calling tasks.

Adversarially Coupled Prompt & Dataset Generator *March'25 - May'25*

Advisor : Anish Nediyanchath (Amazon)

- Developed an **adversarial co-evolution framework** coupling a prompt generator and dataset generator with access to intermediate reasoning steps, achieving a **21× efficiency gain** by producing robust initial prompts and diverse reasoning-annotated datasets, reducing manual effort and accelerating UAT cycles; earned recognition from leadership.

Integration Testing for Stochastic AI systems *May'25 - Aug'25*

Advisor : Anubhav Shrimal (Amazon)

- Developed an **LLM-based, persona-driven multi-turn evaluation framework** for stochastic AI systems, simulating real user behaviors (waiting, task switching, file uploads) and enabled headless CI/CD integration to produce automated pass/fail signals, improving reliability of validation.

AWARE: Adaptive Wear-levelling and Attack Re-mapping Engine *June'24 - Dec'24*

Advisor : Dr Shirshendu Das (IIT-H) | Report |

- Proposed a novel framework that enhances NVM durability and security in LLCs by combining adaptive wear-leveling and attack mitigation through intelligent remapping and workload-aware strategies.

Large Language Models related Hardware Optimizations *Jan'24 - April'24*

Advisors : Dr Rajesh Kedia & Dr Shirshendu Das (IIT-H) | Report |

- Analyzed hardware and software optimizations to improve Large Language Models, identifying gaps like fragmented benchmarking and advocating for unified solutions to boost efficiency.

Achievements

Reliance Foundation Postgraduate Scholarship *'23 - '25*
Awarded to Top 100 Students Nationwide. Scholarship Value: 6 Lakhs

AAAI Travel Grant Awardee *'26*
Competitive grant supporting travel and accommodation to attend AAAI 2026

Amazon's Internal Hackathon *'25*
Top-3. Only Intern Finalist, competing against well-seasoned Applied Scientists & SDEs

TiDeL Hackathon *'24*
First Place | Report |

Amazon Machine Learning Summer School *'24*

Volunteering

Research

Reviewer: AAAI'26, ECCV'22

Sub-Reviewer: CVPR'23, ICLR'21, IJCAI'20, WACV'23, SDM'21

Student Volunteer: AAAI'26, ACML'22, ICML'20

Academic TAships

AI and Emerging Technologies (IIT Hyderabad) ('26, '25, '24, '23, '22)

Deep Learning for Computer Vision (NPTEL) ('24, '20)

Effective Teaching of Machine Learning (CSEDU IIIT Delhi) ('22, '21)

Advanced Topics in Machine Learning (AI 2100 / CS 6360) (IIT Hyderabad) ('21)

References

- Dr. Vineeth N. Balasubramanian, Principal Researcher, Microsoft Research, India | Professor, IIT-H, India.
- Promod Yenigalla, Sr. Applied Science Manager, Amazon, USA.
- Anish Nediyanchath, Sr. Applied Scientist, Amazon, India.
- Anubhav Shrimal, Sr. Applied Scientist, Amazon, USA.